

Finite-Size Scaling of Data-Driven Transfer-Operator Spectra: A Characterization, and an Honest Null for Spectral Domain Extension

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Abstract

Can one predict the long-time statistics of an extensively chaotic field on a large domain by learning a transfer operator only on small domains and extrapolating its spectrum in system size? We study this for the one-dimensional Kuramoto–Sivashinsky (KS) equation. We first establish, as a pure-numeric characterization, that the leading Ruelle–Pollicott / Koopman resonances and the spectral density estimated by delay-embedded EDMD have a *smooth, fittable dependence on domain size L* : a per-wavenumber $1/L$ flow fit on $L \in \{22, 44, 66, 88\}$ predicts the held-out $L = 176$ spectrum to 8.2% (decay rates) and 2.1% (spectral density). Pushed to $L = 1408$ ($64\times$ the base volume) with *zero* large-domain data, the finite-size-scaling (FSS) flow reproduces decay rates to 9.8%, spectral density to 12.5%, two-point correlations to 9.6%, and correlation times to 6.1%. However — and this is the central, pre-registered finding — the FSS machinery is *unnecessary* for KS: the simplest translation-invariant null, interpolating the spectrum of the largest affordable small box ($L = 88$) in wavenumber with no size flow at all, does *better* (3.4% on decay rates), because KS bulk spectra converge to within 4.3% of the thermodynamic limit already by $L = 88$. We report this honestly, quantify the convergence that drives it, and delimit where such a null must fail. Separately, a deep translation- equivariant Koopman autoencoder reproduces the invariant spectral density (8.2%) but not the per-mode resonances (328.3% error on oscillation frequencies), so the data-driven-Koopman (EDMD) operator is our spectral instrument. All numbers are machine-generated and regenerate byte-for- byte.

1 Introduction

Extensively chaotic spatiotemporal systems have a thermodynamic limit: as the domain grows, intensive statistics (Lyapunov-exponent density, correlation lengths, spectral densities) converge, with finite-size corrections that vanish as the box grows [Cross and Hohenberg, 1993, Egolf and Greenside, 1994, Manneville, 1985]. Transfer-operator (Ruelle–Pollicott, RP) resonances [Ruelle, 1986, Pollicott, 1985] encode the statistics that survive to long times in a chaotic system — correlation-decay rates, power spectra, mixing — and are, unlike trajectories, legitimately predictable past the Lyapunov horizon. This paper asks a scaling question about them: *does the size-dependence of a learned operator’s spectrum converge smoothly enough that one can fit it on small domains and extrapolate the statistics of a much larger one, without ever simulating or training at the large size?*

This connects three threads of our own program: transfer-operator statistics as the right target for surrogate models of chaos [Howe, 2026b]; long-horizon statistical (not trajectory) fidelity [Howe, 2026a]; and scaling, rather than brute-force training, as the route to large systems. Two narrowings keep the claim honest. (N1) What is predicted at long horizon is *statistics and slow modes, not trajectories*: $e^{\lambda t}$ spectral continuation cannot beat a positive Lyapunov exponent, but RP resonances do give correlation functions, power spectra, and decay rates [Chekroun et al., 2020]. (N2) Koopman/RP spectral *estimation and forecasting* is an occupied field [Chekroun et al., 2020,

Arbabi and Mezić, 2017, Das and Giannakis, 2019, Williams et al., 2015]; we treat it as substrate. The only mechanism we test as novel is *domain-size extrapolation* of a learned operator’s spectrum via a learned finite-size-scaling flow.

Contributions and honest outcome. (i) A characterization (§4): the finite-size dependence of EDMD-estimated KS RP/Koopman spectra is smooth and $1/L$ -fittable (*GATE S passes*); this is, to our knowledge, the first such measurement. (ii) A working FSS flow (§6) that extrapolates the spectrum to $64\times$ the base volume at single-digit-percent error from zero large-domain data. (iii) The pre-registered null result (§7): because KS spectra converge by $L \approx 4\times$ the base, a no-flow null (interpolate the largest small box) *beats* the flow; the FSS mechanism adds nothing over translation invariance for KS. We report this as the primary finding and delimit where it must fail. (iv) An honest negative for deep operators (§5): a translation-equivariant Koopman autoencoder matches the invariant measure but not per-mode resonances.

2 Related work

RP resonances of stochastic/observed systems. Chekroun et al. [2020], Tantet et al. [2020a,b] develop reduced-state-space RP resonances (with an El-Niño application); our substrate is this machinery, estimated from data. **Koopman spectral estimation and forecasting.** EDMD [Williams et al., 2015], delay-coordinate / Hankel Koopman [Arbabi and Mezić, 2017, Das and Giannakis, 2019, Mezić, 2005, Takens, 1981], DMD for analytic maps [Slipantschuk et al., 2020], and Koopman autoencoders [Lusch et al., 2018, Otto and Rowley, 2019, Azencot et al., 2020] are the fixed-domain tools we build on. **Domain extension for neural operators** is the named adjacent problem: de Villeroché et al. [2026] achieve zero-shot generalization of transformer neural operators to larger domains via attention locality, and Huang et al. [2025] via domain-decomposition operator learning; both are *locality* routes. Ours is a *spectral / renormalization* route, and our finding is that for a fast-converging system the locality-free null already suffices — so we cite-compare rather than reimplement. **RG $\times$ ML:** neural-network renormalization group [Li and Wang, 2018] and analyses of when learning dynamics renormalize [Zhang, 2025] share the RG framing but target sampling / training dynamics, not operator-spectrum domain extension. Our inspiration is Wilsonian momentum-shell scaling; empirically the KS data support only a smooth-in- $1/L$ fit, and we do not oversell the RG language. Dynamic-RG analyses of the *noisy* KS equation place its long-wavelength field in the KPZ universality class [Ueno et al., 2005]; that is a different object (field scaling exponents), not the finite-size dependence of the transfer-operator spectrum we measure.

3 Setup

We integrate KS, $u_t = -uu_x - u_{xx} - u_{xxx}$, periodic on $[0, L)$, with a pseudospectral ETDRK4 scheme [Kassam and Trefethen, 2005] (2/3 dealiasing), vendored and regression-pinned against a prior validated implementation [Howe, 2026b]. The grid resolution $\Delta x = 22/64$ is fixed for all sizes, $dt = 0.25$, samples every 0.5 time units after a 500-unit transient; 3 seeds per size.

Per-sector spectra. Translation invariance block-diagonalizes the dynamics over Fourier wavenumbers $k = 2\pi m/L$. For each retained sector ($0 < k \leq 3$) we form the complex mode series $\hat{u}_m(t)/N$ and estimate its leading resonance by Hankel-EDMD: an 8-delay dictionary at stride s samples, one-step map advancing by s , Grams accumulated in a single streaming pass over $\{1, 8, 64, 512\}$; the stride whose window is closest to $3\tau_e(k)$ (with an anti-aliasing guard) is selected. The leading

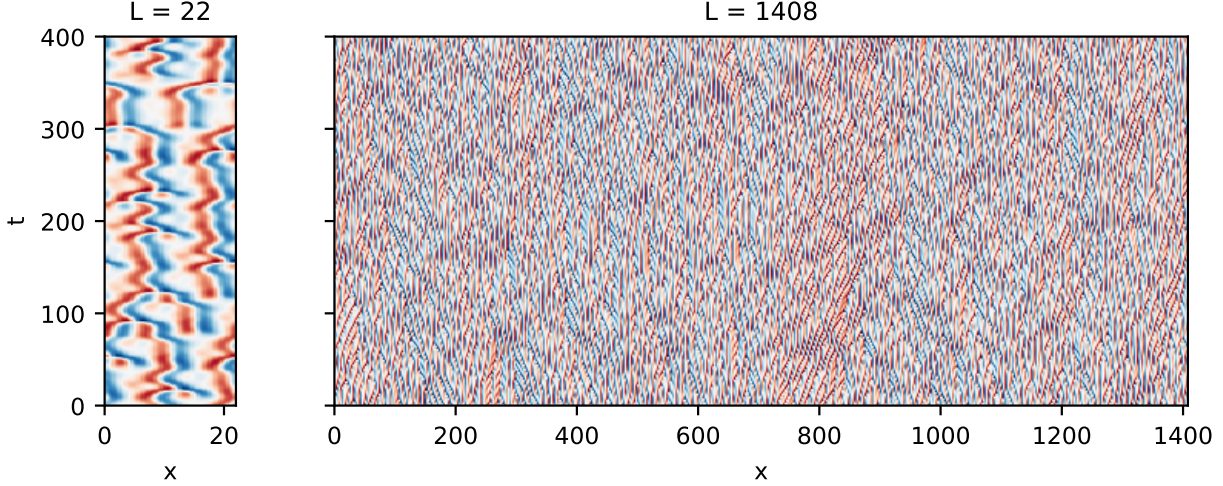


Figure 1: KS space-time fields on the base domain ($L = 22$) and the target ($L = 1408 = 64\times$). The intensive statistics — correlation length, spectral density per unit length — have a thermodynamic limit; this paper asks whether the finite-size dependence of the learned transfer-operator spectrum can be fit on small domains and extrapolated to the large one.

resonance $\lambda(k) = \log \mu / (s \Delta t_s)$ gives decay rate $\gamma(k) = -\text{Re } \lambda$ and frequency $\omega(k) = |\text{Im } \lambda|$. We also accumulate a Welch power spectrum; the spectral density is $\hat{S}(k) = P_m / (2\pi/L)$, the object with a thermodynamic limit. Over all sizes the EDMD-implied autocovariance matches the measured one with median R^2 between 0.94 and 0.99.

4 E0: finite-size scaling of KS spectra exists (GATE S)

On the common grid $k = 2\pi m/22$ (present at every training size) we fit, per wavenumber, $y(L) = y_\infty + c(22/L)^p$ with $p \in \{1, 2\}$ chosen by AICc, seed-averaged and SE-weighted. **GATE S** (pre-registered): a smooth or already-converged size-dependence exists for $\geq 5/7$ wavenumbers (passed: $6/7$ for γ , $7/7$ for $\hat{S}$), and fitting on $\{22, 44, 66, 88\}$ predicts held-out $L = 176$ to $\leq 10\%$ (achieved: 8.2% for γ , 2.1% for $\hat{S}$). **GATE S passes.** Finite-size corrections are large at small L — at $k \approx 0.57$ the decay rate grows from 0.015 ($L = 22$) to 0.087 ($L = 88$), a factor 5.7 — and mostly saturate by $L = 88$; AICc prefers the $1/L^2$ form for $4/7$ of the density points. This smooth, measurable size-dependence is the paper’s foundation (Fig. 2).

5 E1: a deep Koopman operator matches the measure, not the resonances

We train a translation-equivariant convolutional Koopman autoencoder (circular convolutions, a linear circular-convolution propagator, 16 latent channels) at $L = 22$, 11.2 min/model. It reproduces the *invariant spectral density* well — generated fields match the measured density to 8.2% (median) and the two-point correlation to 12.6% — but its per-sector *leading resonances* do not match EDMD: even under generate-and-reestimate (drive the learned stochastic operator, run the identical estimator used on data), the decay rates are off by 86.5% and the oscillation frequencies by 328.3% . The variance-dominated reconstruction loss pins the measure without pinning each

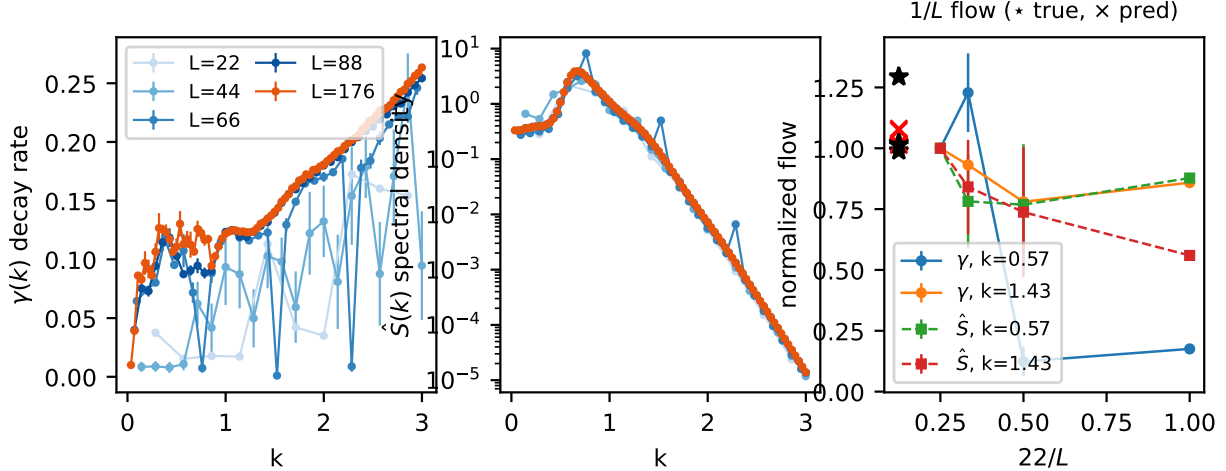


Figure 2: E0 characterization. Left: decay rate $\gamma(k)$; middle: spectral density $\hat{S}(k)$; both vs. wavenumber for $L \in \{22, 44, 66, 88, 176\}$. Right: the per-wavenumber $1/L$ flow at two common wavenumbers, normalized; $\star$ is the held-out $L = 176$ truth and $\times$ the flow prediction. GATE S passes.

mode’s resonant phase. Per our pre-registered kill condition K3, we therefore build the FSS flow on the data-driven Koopman (EDMD) spectrum — itself a learned transfer operator — and report the deep operator as this measure-fidelity/resonance-infidelity result.

6 E2: the finite-size-scaling flow works

The FSS flow generalizes the GATE-S recipe to arbitrary target wavenumbers: for each target k , gather the value at the nearest mode of each training size and fit the seed-averaged $1/L$ (or $1/L^2$, AICc) model, extrapolating to the target L . We report the headline over the *full-support band* $k \in [2\pi/22, 2.2]$ (wavenumbers present at all four training sizes); the partial-support band $[2\pi/88, 2\pi/22]$ and the genuinely new long-wave modes ($k < 2\pi/88$) are reported separately.

Fitting on $\{22, 44, 66, 88\}$ and extrapolating with *zero* large-domain data, the flow predicts the $L = 1408$ spectrum (672 sectors, $64\times$ the base volume): decay rates to 9.8%, spectral density to 12.5%, band-limited two-point correlation to 9.6%, correlation times to 6.1%, and a slow-mode-subspace Jaccard overlap of 0.38 (Fig. 3). The new long-wave band is revealing: its *spectral density* extrapolates almost perfectly (4.1% over 15 modes, since $\hat{S}$ is flat at low k) but its *decay rates* do not (394.8% error) — these hydrodynamic modes are far slower than anything in the training set, so the dynamics, unlike the statics, cannot be extrapolated in wavenumber. Taken alone this is a positive domain-extension result; §7 shows why it is nonetheless unnecessary.

7 E3: the honest null — K2 fires

The house rule is that the null control is the experiment that makes it a paper. We compare the flow against the strongest no-flow null: **interp-88**, which interpolates the measured $L = 88$ spectrum in wavenumber to the target grid with *no size flow at all* (the spectral analog of evaluating a small-domain operator on a larger domain, the locality/zero-shot route [de Villeroché et al., 2026]). Table 1 is decisive: interp-88 predicts the $L = 1408$ decay rates to 3.4%, density to 0.6%, and correlations to 1.8% — *better* than the flow on all 5 headline metrics. Our pre-registered kill

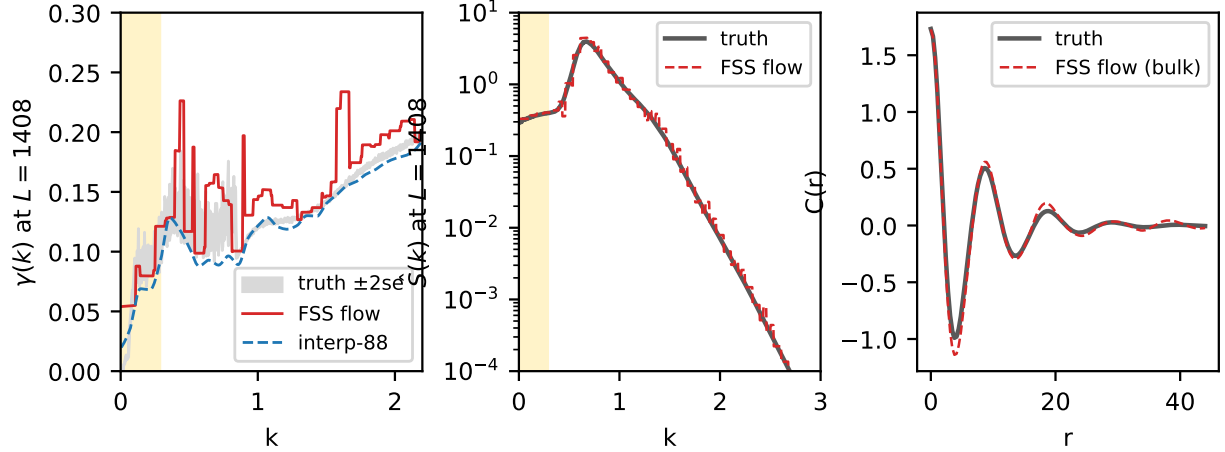


Figure 3: E2 domain extension to $L = 1408$. Decay rate $\gamma(k)$, spectral density $\hat{S}(k)$, and two-point correlation $C(r)$: truth (grey), FSS flow (red), and the no-flow null interp-88 (blue, left). Both are fit only on $L \leq 88$; the shaded band is the sub- $2\pi/22$ region (partial/new-mode support) reported separately.

Table 1: Domain extension to $L = 1408$ ($64\times$ the base volume), headline band $k \in [2\pi/22, 2.2]$. Columns: median relative error of decay rate γ (%), median $|\log_{10}|$ of the density ratio, band-limited $C(r)$ relative L_2 (%), correlation-time τ_e error (%), slow-mode Jaccard overlap. The FSS flow works, but the no-flow null *interp-88* does better on every metric: **K2 fires**.

method	γ	$\hat{S} \log_{10} $	$C(r)$	τ_e	slow
FSS flow (ours)	10.0	0.050	6.6	6.9	0.35
FSS flow, base ≤ 44	87.8	1.202	1560.3	34.2	0.00
interp-88 (null)	3.4	0.002	1.8	4.3	0.44
interp-44 (null)	34.7	0.008	8.6	28.5	0.38
interp-22 (null)	80.4	0.135	38.5	148.2	0.42
EDMD@ L , $T=2000$	46.2	0.014	1.3	69.2	0.27
EDMD@ L , $T=10000$	3.4	0.006	0.9	2.3	0.38
strict tiling (null)	—	—	137.3	—	—

condition **K2 fires** (the flow wins 0/5): the FSS mechanism adds nothing over translation-invariant interpolation for KS.

The mechanism is fast convergence (§8, Table 2): the KS bulk spectrum is within 4.3% (decay rates) and 0.5% ($\hat{S}$) of the $L = 1408$ limit *already at* $L = 88$, so using the largest affordable box directly beats extrapolating a $1/L$ flow, which over-corrects past the already-converged largest size. The weaker strict-tiling null (periodically tile the $L = 22$ field) captures band-integrated power to 52.4% but fails on the two-point correlation (137.3%) through the spurious $L = 22$ periodicity it imposes — a reminder that tiling, unlike smooth k -interpolation, injects artificial long-range order. A flow fit on an aggressive base $\{22, 33, 44\}$ is worse still (87.8% on decay rates; its density extrapolation is numerically unstable), and even there interp-44 wins. With *some* large-domain data, direct EDMD at the target reaches 3.4% (decay rates) from only 10,000 time units — and the full oracle costs $8.0\times$ our small-domain compute, while interp-88 costs the same as the flow.

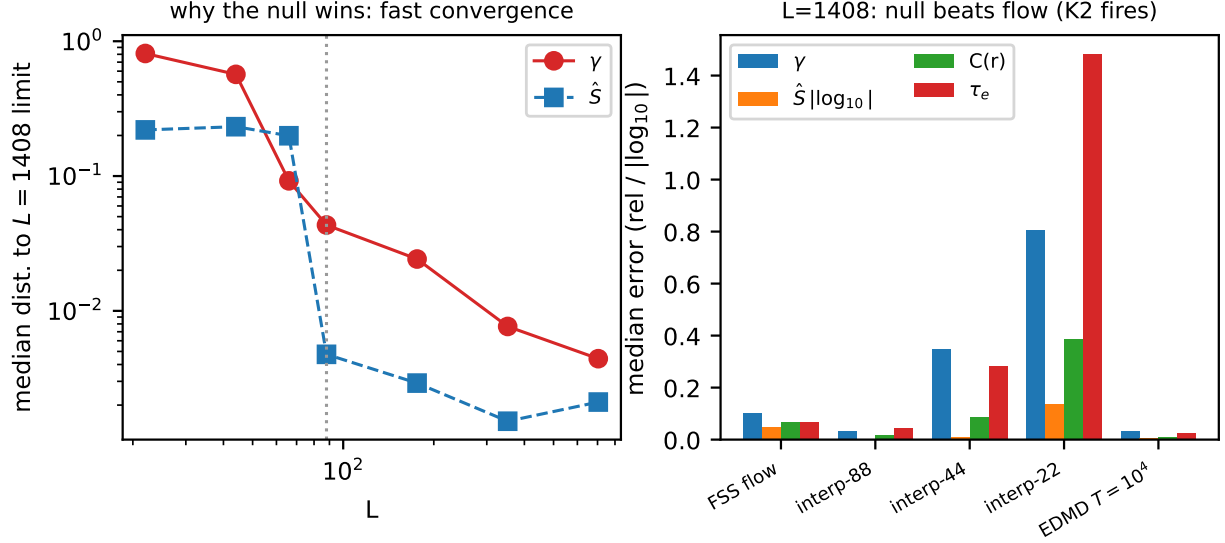


Figure 4: Left: the mechanism — the KS bulk spectrum’s median distance to the $L = 1408$ limit collapses to a few percent by $L = 88$ (dotted line), so the largest small box is already near-converged. Right: at $L = 1408$, the no-flow null interp-88 beats the FSS flow on every headline metric — K2 fires.

Table 2: Convergence of the KS bulk spectrum to the $L = 1408$ limit: median relative distance (%) on the common grid $k = 2\pi m/22$. The decay rates are within a few percent of the limit already by $L = 88$; this fast convergence is why interpolating the largest small box (interp-88) beats the finite-size flow.

L	γ to limit (%)	$\hat{S}$ to limit (%)
22	81.1	22.0
44	56.8	23.2
66	9.2	19.9
88	4.3	0.5
176	2.4	0.3
352	0.8	0.2
704	0.4	0.2

8 E4: boundaries — where each route breaks

Convergence (the mechanism). Table 2 reports the median distance of each size’s common-grid spectrum from the $L = 1408$ limit. The decay rates are within 56.8% by $L = 44$ and 4.3% by $L = 88$; this rapid convergence is exactly why interp-88 suffices and the flow is redundant.

Extrapolation distance. Across the ladder $L \in \{176, \dots, 2816\}$ (Table 3) both the flow and interp-88 stay flat in error (flow 10.0% $\rightarrow$ 9.8% over $1408 \rightarrow 2816$; interp-88 3.4% $\rightarrow$ 3.9%), consistent with a converged bulk spectrum: distance in L is not the limiting factor once past $L \approx 88$.

Boundary conditions. Under odd-parity (Dirichlet-type $u = u_{xx} = 0$) boundaries at $L = 176$, the bulk two-point correlation heals to the periodic prediction within a wall layer of 6.5 units; the periodic FSS-flow prediction reproduces the odd-parity *bulk* correlation to 7.6% and the bulk-point temporal autocorrelation to 12.5%, i.e. the size-scaled periodic spectrum transfers to the bulk of a non-periodic domain.

Scale content. Splitting $L = 1408$ into an energy-containing band ($k \leq 1.5$) and a microscale

Table 3: Extrapolation distance: decay-rate median relative error (%) vs. target L (both routes fit only on $L \leq 88$). Error is flat past $L \approx 88$ for both, consistent with an already-converged bulk spectrum; interp-88 leads throughout.

method	176	352	704	1408	2816
FSS flow	7.3	9.9	10.0	10.0	9.8
interp-88	2.4	2.9	3.4	3.4	3.9

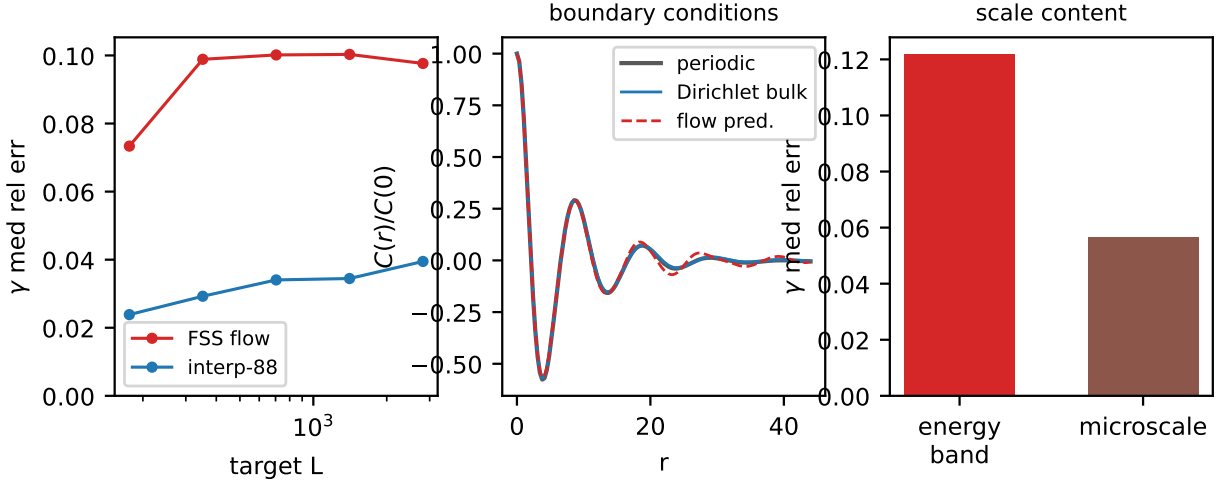


Figure 5: E4. Left: decay-rate error vs. target L (flat past $L \approx 88$ for both routes). Middle: odd-parity (Dirichlet-type) bulk correlation matches the periodic FSS-flow prediction outside a thin wall layer. Right: flow error split by scale band at $L = 1408$.

band ($k > 1.5$), the flow’s decay-rate error is 12.2% in the energy band and 5.7% in the microscale band.

New long-wave modes. The genuinely new modes ($k < 2\pi/88$, absent at every training size) separate statics from dynamics cleanly (Fig. 6): their spectral density extrapolates in wavenumber to 4.1% (the density is flat at low k), but their decay rates, being far slower than anything trained, are not recovered (394.8%). No method — flow or null — predicts the new modes’ *dynamics*; this is the honest ceiling of size-and-wavenumber extrapolation.

9 What we do not claim

We do not claim trajectory forecasting (N1): only statistics and slow modes. We do not claim that the FSS flow beats translation-invariant interpolation for KS — it does not (K2 fires); we claim only that the size-dependence is smooth and fittable, and that both routes extrapolate the bulk statistics to $64\times$ at single-digit-percent error. We make no 2-D/3-D or turbulence claims (none were run). We do not claim the deep operator captures resonances (it does not). We do not claim the new long-wave modes’ *dynamics* are predicted — only their spectral density.

Where the null must fail (proposed, not claimed). interp-88 wins because the KS correlation length is short relative to $L = 88$, so the bulk spectrum is converged. A finite-size flow should be necessary precisely where the largest affordable box is *not* converged: systems with a correlation

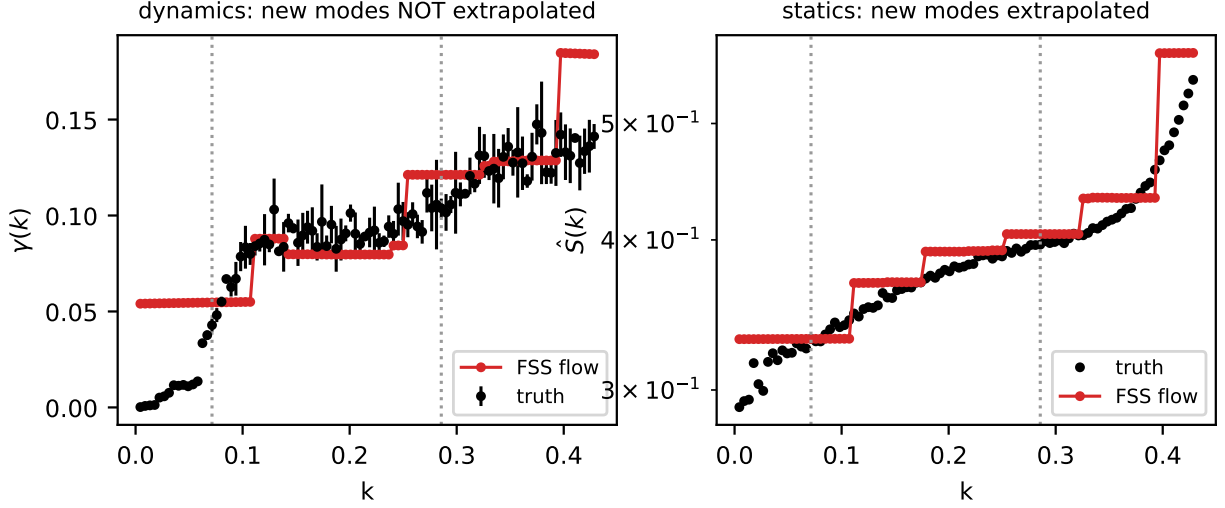


Figure 6: The new long-wave band at $L = 1408$ ($k < 2\pi/22$, dotted lines mark $2\pi/88$ and $2\pi/22$). Left: decay rates of the genuinely new modes are not extrapolated. Right: their spectral density is, because $\hat{S}$ is flat at low k . Statics extrapolate in wavenumber; dynamics do not.

length comparable to the box, long-range coupling, or size-dependent linear instabilities (so that new unstable modes appear as L grows). Testing whether the FSS flow beats interpolation there — where interpolation of a small box must fail — is the natural next experiment.

10 Reproducibility

Every number above is emitted by `gen_paper_numbers.py` from committed JSON and regenerates byte-for-byte (`verify_regen.py`); the integrator, estimators, and scaling fits are unit-tested (spectra vs. analytic linear sectors; integrator regression; scaling round-trip). Simulations use a fixed grid, 3 seeds, and pre-registered gates (PLAN.md). Code and data: DOI to be assigned at submission. Recency re-sweep logged in PLAN.md.

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